



Customer Segmentation Using RFM Metrics

Forming customer clusters and designing differentiated marketing actions for a retail chain — a Business Analytics project leveraging Recency, Frequency, and Monetary analysis.

BUSINESS ANALYTICS PROJECT

RETAIL INTELLIGENCE



The Problem: One-Size-Fits-All Marketing

Retail chains frequently apply uniform marketing strategies across all customers, leading to wasted budgets and missed retention opportunities. A data-driven approach is needed to identify high-value customers, flag those at risk of churning, and re-engage lapsed shoppers.

Inefficient Spend

Broad campaigns dilute impact and inflate cost-per-acquisition.

Missed Retention

High-value customers go unrecognised until they churn.

No Personalisation

Without segmentation, offers fail to resonate with individual behaviour.

What Is RFM Analysis?

RFM scores every customer across three behavioural dimensions, forming the foundation for clustering and targeted marketing.



Recency (R)

Days since the customer's most recent purchase. Recent buyers are more likely to respond to new offers.



Frequency (F)

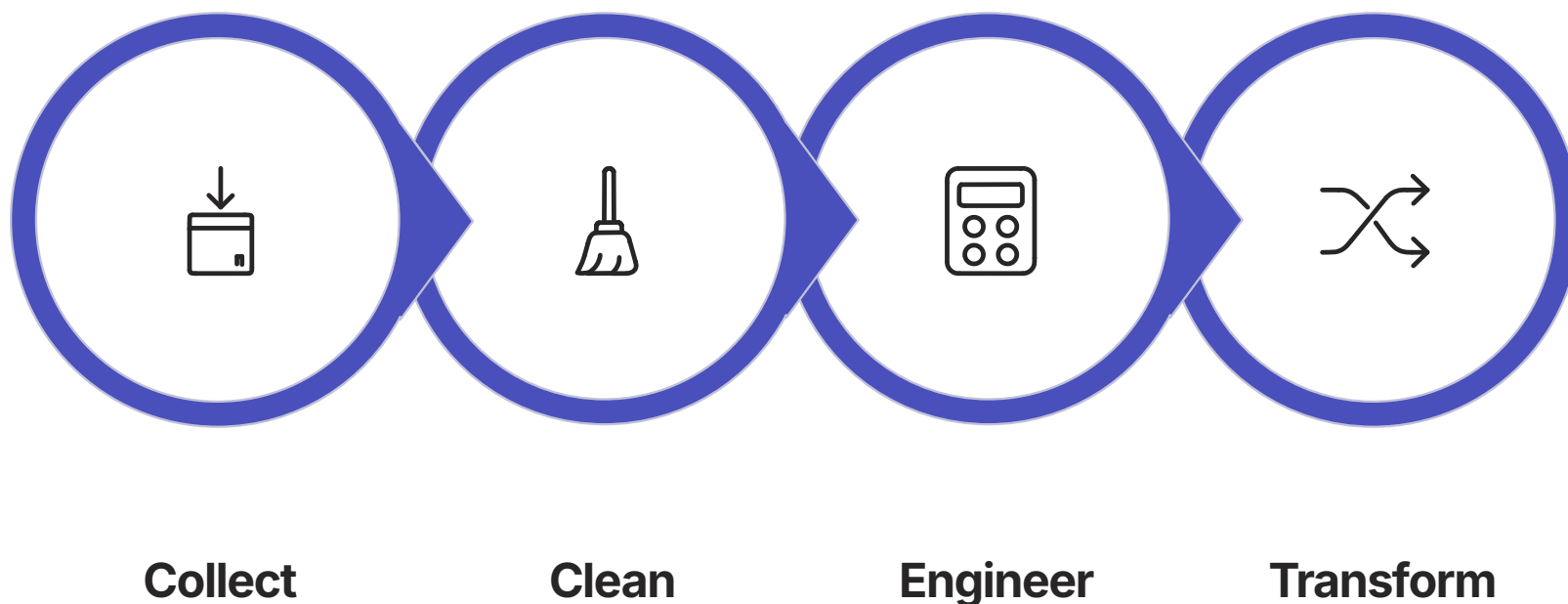
Total number of transactions within the analysis period. Frequent buyers show stronger brand affinity.



Monetary (M)

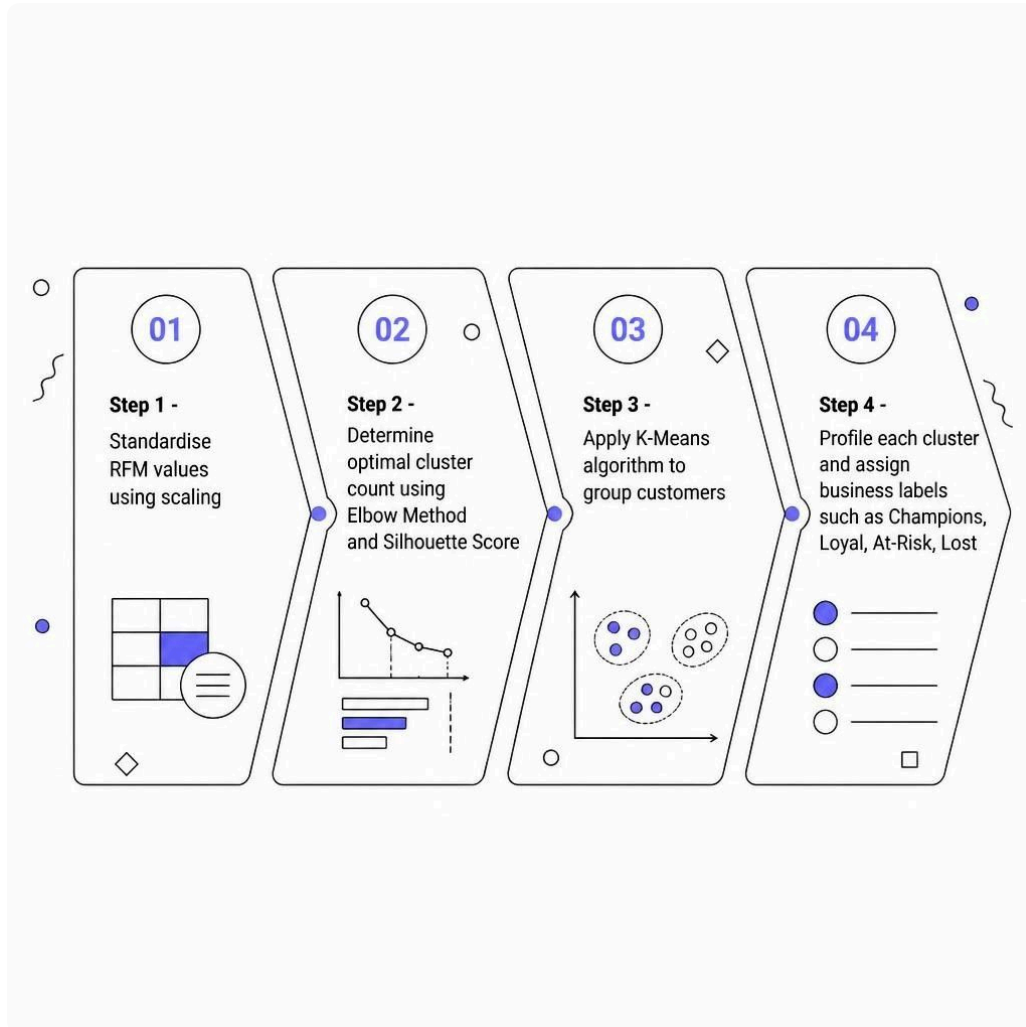
Total or average spend over the analysis period. High spenders represent the greatest revenue potential.

Data Pipeline: From Raw Transactions to RFM Scores



RFM values are log-transformed and standardised before clustering so no single metric dominates the distance calculation. The dataset typically spans 5,000–10,000 transactions across 1,000–2,000 unique customers.

K-Means Clustering: Building Actionable Segments



Why K-Means?

K-Means is efficient, interpretable, and widely adopted for retail segmentation. The optimal cluster count is validated using the **Elbow Method** and **Silhouette Score**, ensuring segments are well-separated and business-meaningful.

- Cluster stability is verified by running the algorithm with multiple initialisations and comparing results.

Customer Segments & Differentiated Marketing Actions

Segment	RFM Profile	Recommended Action
Champions	Low Recency · High F · High M	Loyalty perks, early access, premium offers
Loyal Customers	Low-Moderate R · High F	Upsell, cross-sell, loyalty programme enrolment
Potential Loyalists	Low R · Moderate F	Personalised recommendations, membership incentives
New Customers	Low R · Low F	Welcome offers, onboarding campaigns
At-Risk	High R · Previously High F/M	Win-back offers, personalised discounts, re-engagement emails
Lost Customers	High R · Low F · Low M	Low-cost reactivation or exclude from high-cost spend

Key Business Insights

Revenue is Concentrated

A small Champion segment contributes a disproportionately large share of total revenue — retention here yields the highest ROI.

Churn Signals Are Detectable Early

Rising Recency combined with declining Frequency signals at-risk customers before they fully churn, enabling timely intervention.

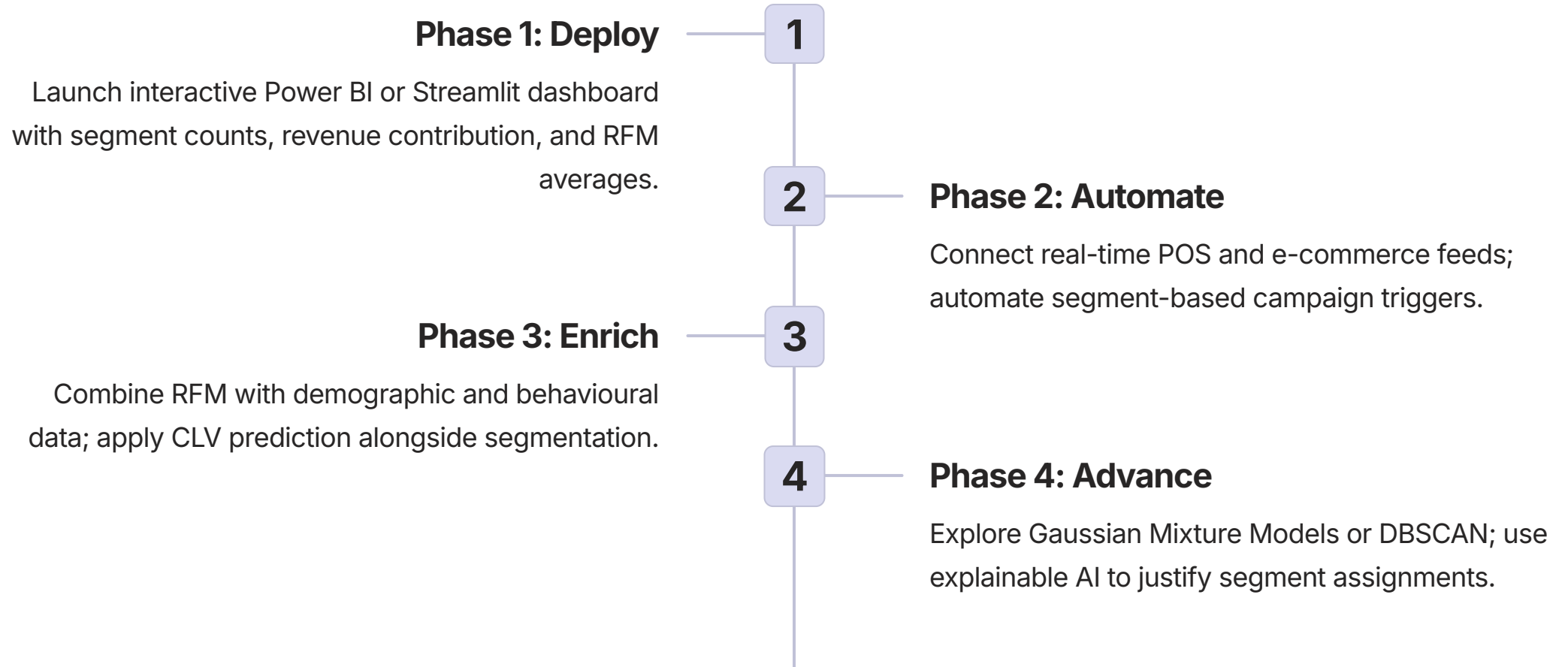
Channel-Level Patterns Emerge

Certain store locations or e-commerce channels concentrate high-value customers, guiding where to invest in personalised outreach.

Automation Scales Low-Value Segments

Large infrequent-customer groups are best served by automated, cost-efficient campaigns rather than high-touch personalised outreach.

Implementation Roadmap & Future Enhancements



📄 **Tools used:** Python, Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn, Power BI / Streamlit, Jupyter / Google Colab, GitHub.